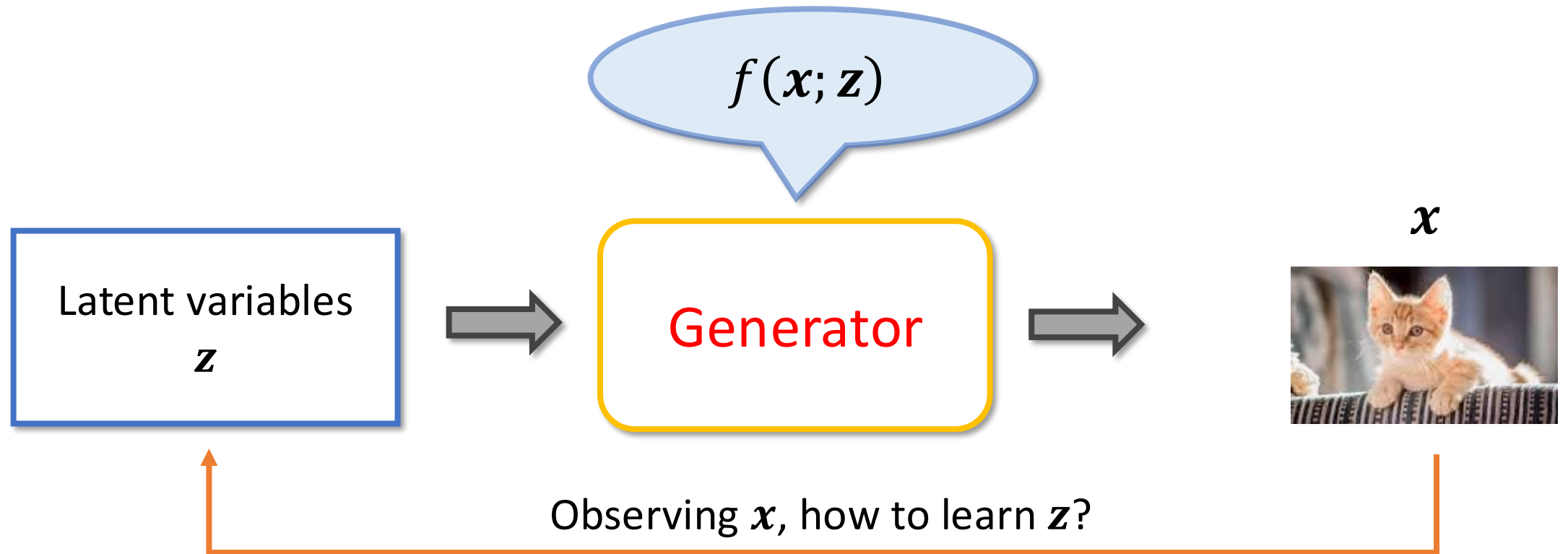


CS 57800 – Statistical Machine Learning

Lecture 2.5 Latent Variables Learning

https://ruizhezhang.com/course_fall_2026.html

Latent Variable Models



Outline

- Matrix Completion
- Method of Moments

Matrix Completion

- $M \in \mathbb{R}^{n_1 \times n_2}$ is an **unknown** matrix
- **Input:** $\{(i_k, j_k), M_{i_k j_k}\}$ for $k \in [m]$
- **Goal:** reconstruct M

This seems to be a silly question, since changing an unobserved entry leaves all data unchanged

- We need a **relation** among entries

5	?	?	2	?
?	4	?	?	3
2	?	5	?	?
?	?	4	?	1

Matrix Completion

Low-rankness $\Leftrightarrow$ Shared latent features

- $M_{ij} = \mathbf{a}_i^\top \mathbf{b}_j$ for $\mathbf{a}_i, \mathbf{b}_j \in \mathbb{R}^r$, $(i, j) \in [n_1] \times [n_2] \Leftrightarrow \text{rank}(M) \leq r \ll \min\{n_1, n_2\}$

$$\begin{array}{ccc} \boxed{M} & = & \boxed{L} \boxed{R^\top} \\ n_1 \times n_2 & & n_1 \times r \quad r \times n_2 \end{array}$$

- Under this assumption, there are only $(n_1 + n_2)r$ hidden parameters, much smaller than $n_1 n_2$
- One observation constrains factors that are shared by many other entries

Matrix Completion

Motivation: Netflix Prize



Leaderboard

Showing Test Score. [Click here to show quiz score](#)

Display top leaders.

Rank	Team Name	Best Test Score	% Improvement	Best Submit Time
Grand Prize - RMSE = 0.8567 - Winning Team: BellKor's Pragmatic Chaos				
1	BellKor's Pragmatic Chaos	0.8567	10.06	2009-07-26 18:18:28
2	The Ensemble	0.8567	10.06	2009-07-26 18:38:22
3	Grand Prize Team	0.8582	9.90	2009-07-10 21:24:40
4	Opera Solutions and Vandelay United	0.8588	9.84	2009-07-10 01:12:31
5	Vandelay Industries !	0.8591	9.81	2009-07-10 00:32:20
6	PragmaticTheory	0.8594	9.77	2009-06-24 12:06:56
7	BellKor in BigChaos	0.8601	9.70	2009-05-13 08:14:09
8	Dace	0.8612	9.59	2009-07-24 17:18:43

	movie 1	movie 2	movie 3	movie 4	movie 5	movie 6	movie 7	movie 8	movie 9	movie 10	...	movie 17770
user 1			1		2							3
user 2		2		3	3			4				
user 3							5	3		4		
user 4	2				3			2				2
user 5		4				5			3			4
user 6			2									
user 7			2					4	2	3		
user 8	3	4				4						
user 9									3			
user 10			1		2							2
...												
user 480189		4			3			3				

Matrix Completion

Rank-1 example:

$$\begin{pmatrix} 2 & 3 & ? \\ 4 & ? & 10 \end{pmatrix}$$

- What if only the first row were observed?

Matrix Completion

Data generation

- $(i_k, j_k) \sim \text{Unif}([n_1] \times [n_2])$ independently for $k \in [m]$
- Sampling is **with replacement**: a location may occur multiple times

Prediction:

$$\mathcal{R}(\hat{X}) = \frac{1}{n_1 n_2} \sum_{ij} |\hat{X}_{ij} - M_{ij}|$$

- Sample complexity: $\sim \varepsilon^{-2} B^2 r n \log n$

Exact recovery:

$$\hat{X} = M$$

- Sample complexity: $\sim r n \log^2 n$

Matrix Completion

Let $M \in \mathbb{R}^{n_1 \times n_2}$ be an **unknown** matrix of **rank $\leq r$** . Given observations of uniformly random entries $\{(i_k, j_k), M_{i_k j_k}\}$ for $k \in [m]$, the goal is to output a matrix $\hat{X} \approx M$

Can we optimize over the low-rank matrices?

$$\min_{\hat{X} \in \mathbb{R}^{n_1 \times n_2} : \text{rank}(\hat{X}) \leq r} \sum_{k=1}^m (\hat{X}_{i_k j_k} - M_{i_k j_k})^2$$

- The set of rank- r matrices are not convex:

$$\frac{1}{2} \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} + \frac{1}{2} \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} = \frac{1}{2} I_2$$

Can we relax to all matrices in $\mathbb{R}^{n_1 \times n_2}$?

- The exact minimizer will set every missing entry to zero

Matrix Completion

The **singular value decomposition (SVD)** of X is

$$X = \sum_{i=1}^s \sigma_i \mathbf{u}_i \mathbf{v}_i^\top, \quad \sigma_i > 0$$

- Rank

$$\text{rank}(X) = |\{i \in [s] : \sigma_i > 0\}| = \|\boldsymbol{\sigma}\|_0$$

- Operator norm

$$\|X\| = \max_{i \in [s]} \sigma_i = \|\boldsymbol{\sigma}\|_\infty$$

- Frobenius norm

$$\|X\|_F = \left(\sum_{i=1}^s \sigma_i^2 \right)^{1/2} = \|\boldsymbol{\sigma}\|_2$$

- Nuclear norm

$$\|X\|_* = \sum_{i=1}^s \sigma_i = \|\boldsymbol{\sigma}\|_1$$

Matrix Completion

Nuclear norm

$$\|X\|_* = \sum_{i=1}^s \sigma_i = \|\boldsymbol{\sigma}\|_1$$

- If $\text{rank}(X) \leq r$, then

$$\|X\|_* \leq \sqrt{r} \left(\sum_{i=1}^s \sigma_i^2 \right)^{1/2} = \sqrt{r} \|X\|_F$$

- If we also know that $\|X\|_{\max} := \max_{ij} |X_{ij}| \leq B$, then

$$\|X\|_* \leq \sqrt{r} \|X\|_F \leq B \sqrt{r n_1 n_2}$$

- Let $\tau := B \sqrt{r n_1 n_2}$.
- The set $\{X \in \mathbb{R}^{n_1 \times n_2} : \|X\|_* \leq \tau\}$ contains all rank $\leq r$ matrices (and some higher rank matrices)

Matrix Completion

Lemma (Nuclear norm/operator norm duality). For every real matrix X ,

$$\|X\|_* = \sup_{Y: \|Y\| \leq 1} \langle Y, X \rangle, \quad \|X\| = \sup_{Y: \|Y\|_* \leq 1} \langle Y, X \rangle$$

where $\langle Y, X \rangle := \text{tr}[Y^\top X] = \sum_{ij} X_{ij} Y_{ij}$

Proof.

- Write $X = \sum_{i=1}^s \sigma_i \mathbf{u}_i \mathbf{v}_i^\top$. If $\|Y\| \leq 1$, then

$$\langle Y, X \rangle = \sum_{i=1}^s \sigma_i \mathbf{u}_i^\top Y \mathbf{v}_i \leq \sum_{i=1}^s \sigma_i \|\mathbf{u}_i\| \|Y \mathbf{v}_i\| \leq \sum_{i=1}^s \sigma_i = \|X\|_*$$

- Equality is attained by $Y = \sum_{i=1}^s \mathbf{u}_i \mathbf{v}_i^\top$



Matrix Completion

Lemma (Nuclear norm/operator norm duality). For every real matrix X ,

$$\|X\|_* = \sup_{Y: \|Y\| \leq 1} \langle Y, X \rangle, \quad \|X\| = \sup_{Y: \|Y\|_* \leq 1} \langle Y, X \rangle$$

where $\langle Y, X \rangle := \text{tr}[Y^\top X] = \sum_{ij} X_{ij} Y_{ij}$

- The nuclear norm can be written as a supremum of linear functions of X , which is **convex**

$$\hat{X} = \arg \min_X \|X\|_* \quad \text{s. t.} \quad X_{i_k j_k} = M_{i_k j_k} \quad \forall k \in [m], \quad \|X\|_{\max} \leq B$$

- Convex optimization, and M is a feasible solution so $\|\hat{X}\|_* \leq \|M\|_*$
- **Generalization error?**

Matrix Completion

Two more facts about the nuclear norm

Fact 1. Let $\mathcal{A} = \{uv^\top : \|u\| \leq 1, \|v\| \leq 1\}$. Then
$$\{X : \|X\|_* \leq 1\} = \text{conv}(\mathcal{A})$$

Fact 2. For every X ,

$$\|X\|_* = \min_{X=LR^\top} \frac{1}{2} (\|L\|_F^2 + \|R\|_F^2)$$

- The nuclear norm minimization can be viewed as a **least-square on the latent factors**

Matrix Completion

Learning theory for matrix completion

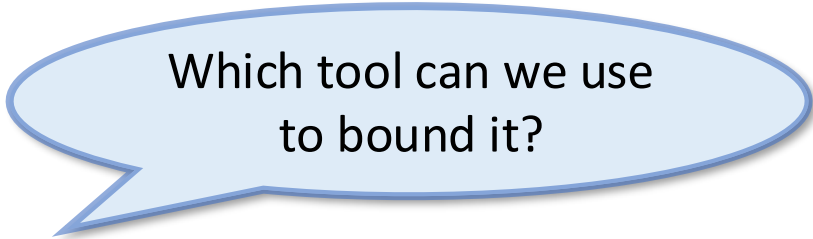
- Domain $\Omega = [n_1] \times [n_2]$
- Data generation distribution $\mathcal{D}$ uniformly samples $(i, j) \in \Omega$
- Hypothesis class

$$\mathcal{H}_{\tau, B} = \{X : \|X\|_* \leq \tau, \|X\|_{\max} \leq B\}$$

- Loss function $\ell(a, b) = |a - b|$
- Generalization error

$$\mathcal{R}(X) = \mathbb{E}_{\mathcal{D}}[|X_{ij} - M_{ij}|]$$

- Rademacher complexity!



Which tool can we use to bound it?

Matrix Completion

Theorem. Assume that the loss function ℓ is bounded by B . Then, with probability $1 - \delta$,

$$\mathcal{R}(\text{ERM}_{\mathcal{H}}(S)) - \inf_{h \in \mathcal{H}} \mathcal{R}(h) \lesssim \mathbb{E}_{S' \sim \mathcal{D}^m} [\mathcal{R}(\ell \circ \mathcal{H} \circ S')] + B \sqrt{\frac{\log(1/\delta)}{m}}$$

- In our case,

$$\hat{X} = \arg \min_X \|X\|_* \quad \text{s.t.} \quad X_{i_k j_k} = M_{i_k j_k} \quad \forall k \in [m], \quad \|X\|_{\max} \leq B$$

- $\hat{X}$ is an ERM learner for $\mathcal{H}_{\tau, B}$
- $\inf_{X \in \mathcal{H}_{\tau, B}} \mathcal{R}(X) = 0$ since M is a feasible solution
- It suffices to bound $\mathbb{E}_{S' \sim \mathcal{D}^m} [\mathcal{R}(\ell \circ \mathcal{H}_{\tau, B} \circ S')]$

Matrix Completion

$$\mathbb{E}_{S' \sim \mathcal{D}^m} [\mathbf{R}(\ell \circ \mathcal{H}_{\tau, B} \circ S')]$$

- Let $S' = \{(i_1, j_1), \dots, (i_m, j_m)\}$
- $\mathcal{H}_{\tau, B} \circ S' = \{(X_{i_1 j_1}, \dots, X_{i_m j_m}) : X \in \mathcal{H}_{\tau, B}\}$
- $\ell \circ \mathcal{H}_{\tau, B} \circ S' = \{(|X_{i_1 j_1} - M_{i_1 j_1}|, \dots, |X_{i_m j_m} - M_{i_m j_m}|) : X \in \mathcal{H}_{\tau, B}\}$
- Its Rademacher complexity is

$$\mathbf{R}(\ell \circ \mathcal{H}_{\tau, B} \circ S') = \frac{1}{m} \mathbb{E}_{\omega \sim \{\pm 1\}^m} \left[\sup_{X \in \mathcal{H}_{\tau, B}} \sum_{k=1}^m \omega_k |X_{i_k j_k} - M_{i_k j_k}| \right]$$

Matrix Completion

Contraction lemma. For each $i \in [m]$, let $\phi_i : \mathbb{R} \rightarrow \mathbb{R}$ be ρ -Lipschitz. For $A \subset \mathbb{R}^m$, let $\phi \circ A := \{(\phi_1(a_1), \dots, \phi_m(a_m)) \mid a \in A\} \subset \mathbb{R}^m$. Then,

$$R(\phi \circ A) \leq \rho \cdot R(A)$$

- $\ell_k(x) = |x - M_{i_k j_k}|$ is 1-Lipschitz

$$|\ell_k(x) - \ell_k(y)| \leq |(x - M_{i_k j_k}) - (y - M_{i_k j_k})| = |x - y|$$

- Thus,

$$R(\ell \circ \mathcal{H}_{\tau, B} \circ S') \leq R(\mathcal{H}_{\tau, B} \circ S') = \frac{1}{m} \mathbb{E}_{\omega \sim \{\pm 1\}^m} \left[\sup_{X \in \mathcal{H}_{\tau, B}} \sum_{k=1}^m \omega_k X_{i_k j_k} \right]$$

Matrix Completion

- Let

$$E_{ij} = \mathbf{e}_i \mathbf{e}_j^\top, \quad \forall ij \in [n_1] \times [n_2]$$

be an orthonormal basis of $\mathbb{R}^{n_1 \times n_2}$, and $\langle X, E_{ij} \rangle = X_{ij}$

- For $\omega \in \{-1, 1\}^m$, define

$$\Sigma_\omega = \sum_{k=1}^m \omega_k E_{i_k j_k}$$

- Then,

$$\sup_{X \in \mathcal{H}_{\tau, B}} \sum_{k=1}^m \omega_k X_{i_k j_k} = \sup_{X \in \mathcal{H}_{\tau, B}} \langle X, \Sigma_\omega \rangle \leq \sup_{X: \|X\|_* \leq \tau} \langle X, \Sigma_\omega \rangle = \tau \|\Sigma_\omega\|$$

Matrix Completion

- Thus,

$$R(\ell \circ \mathcal{H}_{\tau,B} \circ S') \leq \frac{\tau}{m} \mathbb{E}_{\omega \sim \{\pm 1\}^m} [\|\Sigma_\omega\|]$$

and

$$\mathbb{E}_{S' \sim \mathcal{D}^m} [R(\ell \circ \mathcal{H}_{\tau,B} \circ S')] \leq \frac{\tau}{m} \mathbb{E}_{S', \omega} [\|\Sigma_\omega\|]$$

- Σ_ω is a **random matrix**, and we want to bound its operator norm

$$\mathbb{E}_{S', \omega} [\|\Sigma_\omega\|] = \mathbb{E}_{\{i_k j_k\}, \omega} \left[\left\| \sum_{k=1}^m \omega_k E_{i_k j_k} \right\| \right]$$

Matrix Completion

Theorem (Rectangular matrix Bernstein inequality). Let $Z_1, \dots, Z_m \in \mathbb{R}^{n_1 \times n_2}$ be independent random matrices such that $\mathbb{E}[Z_k] = 0$ and $\|Z_k\| \leq L$ almost surely. Define the matrix variance proxy

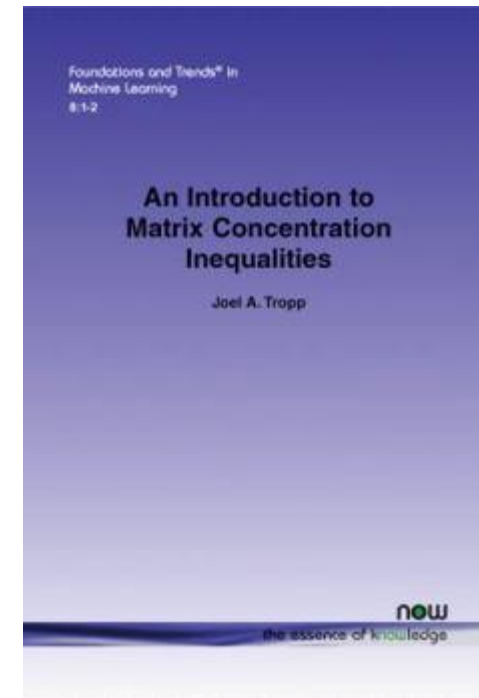
$$v = \max \left\{ \left\| \sum_k \mathbb{E}[Z_k Z_k^\top] \right\|, \left\| \sum_k \mathbb{E}[Z_k^\top Z_k] \right\| \right\}$$

For any $t \geq 0$,

$$\Pr \left[\left\| \sum_k Z_k \right\| \geq t \right] \leq (n_1 + n_2) \exp \left(-\frac{t^2}{2(v + Lt/3)} \right),$$

and

$$\mathbb{E} \left[\left\| \sum_k Z_k \right\| \right] \leq \sqrt{2v \log(n_1 + n_2)} + \frac{L}{3} \log(n_1 + n_2)$$



Matrix Completion

$$\mathbb{E}_{\{i_k j_k\}, \omega} \left[\left\| \sum_{k=1}^m \omega_k E_{i_k j_k} \right\| \right]$$

- Let $Z_k = \omega_k E_{i_k j_k}$ for $k \in [m]$.
- $\mathbb{E}[Z_k] = 0, L = 1,$

$$\mathbb{E}[Z_k Z_k^\top] = \mathbb{E}[E_{i_k j_k} E_{i_k j_k}^\top] = \mathbb{E}[\mathbf{e}_{i_k} \mathbf{e}_{i_k}^\top] = \frac{1}{n_1} I_{n_1}, \quad \mathbb{E}[Z_k^\top Z_k] = \frac{1}{n_2} I_{n_2}$$

- $v = \max\left\{\frac{m}{n_1}, \frac{m}{n_2}\right\} = \frac{m}{n_1}$ if we assume $n_1 \leq n_2$
-

$$\mathbb{E} \left[\left\| \sum_k Z_k \right\| \right] \leq \sqrt{\frac{2m}{n_1} \log(n_1 + n_2)} + \frac{1}{3} \log(n_1 + n_2)$$

Matrix Completion

Theorem. Suppose $\|M\|_* \leq \tau$ and $\|M\|_{\max} \leq B$. Given m independent uniform entries of M . Let $\hat{X}$ be the solution of the optimization problem:

$$\hat{X} = \arg \min_X \|X\|_* \quad \text{s.t.} \quad X_{i_k j_k} = M_{i_k j_k} \quad \forall k \in [m], \quad \|X\|_{\max} \leq B$$

With probability at least $1 - \delta$,

$$\frac{1}{n_1 n_2} \sum_{ij} |\hat{X}_{ij} - M_{ij}| \lesssim \tau \left(\sqrt{\frac{\log(n_1 + n_2)}{m n_1}} + \frac{\log(n_1 + n_2)}{m} \right) + B \sqrt{\frac{\log(1/\delta)}{m}}$$

- If $\text{rank}(M) \leq r$, then $\tau \leq B\sqrt{r n_1 n_2}$, and we get that

$$\frac{1}{n_1 n_2} \sum_{ij} |\hat{X}_{ij} - M_{ij}| \lesssim B \sqrt{\frac{r n_2 \log(n_1 + n_2)}{m}} + B \sqrt{\frac{\log(1/\delta)}{m}}$$

Matrix Completion

Theorem. Suppose $\|M\|_* \leq \tau$ and $\|M\|_{\max} \leq B$. Given m independent uniform entries of M . Let $\hat{X}$ be the solution of the optimization problem:

$$\hat{X} = \arg \min_X \|X\|_* \quad \text{s.t.} \quad X_{i_k j_k} = M_{i_k j_k} \quad \forall k \in [m], \quad \|X\|_{\max} \leq B$$

With probability at least $1 - \delta$,

$$\frac{1}{n_1 n_2} \sum_{ij} |\hat{X}_{ij} - M_{ij}| \lesssim \tau \left(\sqrt{\frac{\log(n_1 + n_2)}{m n_1}} + \frac{\log(n_1 + n_2)}{m} \right) + B \sqrt{\frac{\log(1/\delta)}{m}}$$

- To achieve mean-error at most ε , it suffices to take

$$m = \Omega \left(\frac{B^2}{\varepsilon^2} (r n_2 \log(n_1 + n_2) + \log(1/\delta)) \right)$$

Matrix Completion

Theorem. Suppose $\|M\|_* \leq \tau$ and $\|M\|_{\max} \leq B$. Given m independent uniform entries of M . Let $\hat{X}$ be the solution of the optimization problem:

$$\hat{X} = \arg \min_X \|X\|_* \quad \text{s.t.} \quad X_{i_k j_k} = M_{i_k j_k} \quad \forall k \in [m], \quad \|X\|_{\max} \leq B$$

With probability at least $1 - \delta$,

$$\frac{1}{n_1 n_2} \sum_{ij} |\hat{X}_{ij} - M_{ij}| \lesssim \tau \left(\sqrt{\frac{\log(n_1 + n_2)}{m n_1}} + \frac{\log(n_1 + n_2)}{m} \right) + B \sqrt{\frac{\log(1/\delta)}{m}}$$

- This generalization bound still holds (with minor changes) if the observation entries are **noisy**

$$\frac{1}{m} \sum_{k=1}^m |X_{i_k j_k} - M_{i_k j_k}| \leq \eta \quad \Rightarrow \quad \frac{1}{n_1 n_2} \sum_{ij} |\hat{X}_{ij} - M_{ij}| \leq \text{RHS} + \eta$$

Matrix Completion

$$\hat{X} = \arg \min_X \|X\|_* \quad \text{s.t.} \quad X_{i_k j_k} = M_{i_k j_k} \quad \forall k \in [m]$$

Can the solution exactly recover M ?

- Consider $M = \mathbf{e}_1 \mathbf{e}_1^\top = \begin{pmatrix} 1 & 0 & \cdots & 0 \\ 0 & 0 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 0 \end{pmatrix}$
- Unless we observe M_{11} , the data are identical to those from the all-zero matrix

$$\Pr[M_{11} \text{ is missed}] = \left(1 - \frac{1}{n^2}\right)^m$$

- Yet predicting zero has mean absolute error $1/n^2$

Matrix Completion

The **coherence** μ of a subspace $U \subseteq \mathbb{R}^n$ of dimension r is

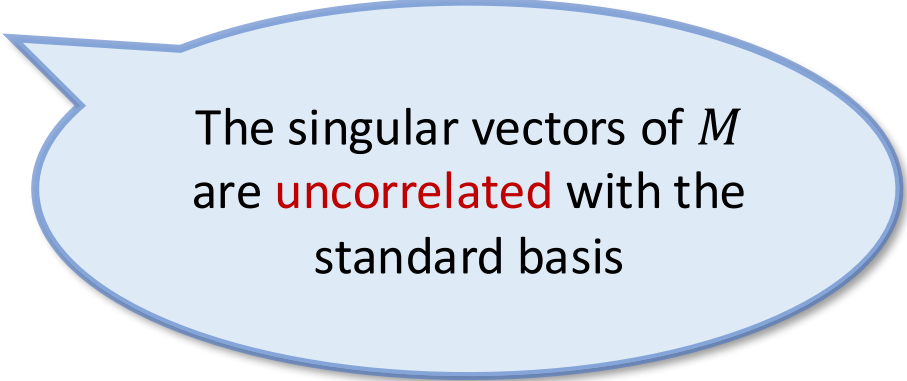
$$\mu = \frac{n}{r} \max_i \|P_U \mathbf{e}_i\|_2^2,$$

where P_U is the orthogonal projection onto U

- If U is a uniformly random subspace, then $\mu = \tilde{O}(1)$
- If $U = \mathbf{e}_1$, then $\mu = n/r$

Let $n_1 = n_2 = n$ for simplicity and let $M = U\Sigma V^\top$ be the SVD with $U, V \in \mathbb{R}^{n \times r}, \Sigma \in \mathbb{R}^{r \times r}$

- We assume that $\mu(U), \mu(V) \leq \mu_0$
- We also assume that $\|UV^\top\|_{\max} \leq \mu_1 \sqrt{r}/n$



The singular vectors of M
are **uncorrelated** with the
standard basis

Matrix Completion

Let $n_1 = n_2 = n$ for simplicity and let $M = U\Sigma V^\top$ be the SVD with $U, V \in \mathbb{R}^{n \times r}, \Sigma \in \mathbb{R}^{r \times r}$

- We assume that $\mu(U), \mu(V) \leq \mu_0$
- We also assume that $\|UV^\top\|_{\max} \leq \mu_1\sqrt{r}/n$

Theorem (Exact Recovery). Under these assumptions, with probability $1 - \delta$, the nuclear norm minimization

$$\hat{X} = \arg \min_X \|X\|_* \quad \text{s. t.} \quad X_{i_k j_k} = M_{i_k j_k} \quad \forall k \in [m]$$

will exactly recover M if

$$m \geq \Omega(\max\{\mu_1^2, \mu_0\} rn \log^2(n/\delta))$$

Matrix Completion

High-level approach

- Suppose $M + Z$ is another candidate solution of the nuclear norm minimization
- We will find a **certificate** matrix B such that
 - $\|B\| \leq 1$
 - $\|M + Z\|_* \geq \langle M + Z, B \rangle > \|M\|_*$
- Thus, $M + Z$ is not optimal

Matrix Completion

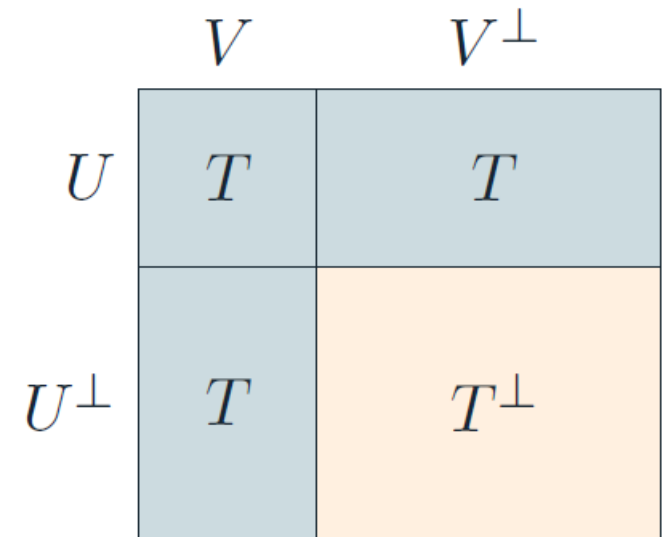
- Recall $M = U\Sigma V^\top$. Let $\mathbf{u}_1, \dots, \mathbf{u}_r$ be the columns of U and $\mathbf{v}_1, \dots, \mathbf{v}_r$ be the columns of V
- We can extend then to orthonormal bases of $\mathbb{R}^n$ with $\mathbf{u}_{r+1}, \dots, \mathbf{u}_n$ and $\mathbf{v}_{r+1}, \dots, \mathbf{v}_n$, respectively
- Define a linear subspace of matrices:

$$T = \text{span}\{\mathbf{u}_i \mathbf{v}_j^\top : i \in [r] \text{ or } j \in [r]\}$$

Then

$$T^\perp = \text{span}\{\mathbf{u}_i \mathbf{v}_j^\top : r+1 \leq i, j \leq n\}$$

- $\dim(T) = r^2 + 2r(n-r)$
- $\mathcal{P}_T[Z] = P_U Z + Z P_V - P_U Z P_V$
- $\dim(T^\perp) = (n-r)^2$
- $\mathcal{P}_{T^\perp}[Z] = P_{U^\perp} Z P_{V^\perp}$



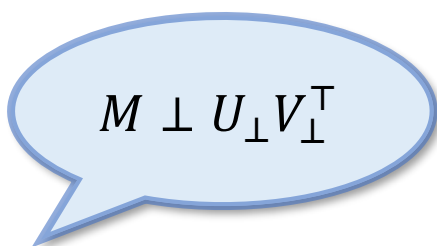
Matrix Completion

- Let Ω be the locations of the observations. Our goal is to show that $\|M + Z\|_* > \|M\|_*$ for any Z supported on Ω^c by finding a matrix B such that $\langle M + Z, B \rangle > \|M\|_*$
- Let $U_\perp, V_\perp$ be the left- and right-singular vectors of $\mathcal{P}_{T^\perp}[Z]$, and define

$$B = [U \quad U_\perp] \begin{bmatrix} V^\top \\ V_\perp^\top \end{bmatrix} = UV^\top + U_\perp V_\perp^\top$$

- $\|B\| \leq 1$ since the above expression is an SVD for B

$$\begin{aligned} \|M + Z\|_* &\geq \langle M + Z, B \rangle = \langle M + Z, UV^\top + U_\perp V_\perp^\top \rangle \\ &= \langle M, UV^\top \rangle + \langle Z, UV^\top + U_\perp V_\perp^\top \rangle \\ &= \|M\|_* + \langle Z, UV^\top + U_\perp V_\perp^\top \rangle \end{aligned}$$


$$M \perp U_\perp V_\perp^\top$$

We need to show that $\langle Z, UV^\top + U_\perp V_\perp^\top \rangle > 0$ for any Z supported on Ω^c

Matrix Completion

Suppose there is a **helper matrix** Y that satisfies the following conditions:

i. Y is supported on Ω

$$\langle Y, Z \rangle = 0$$

ii. $\|\mathcal{P}_T[Y] - UV^\top\|_F \leq \sqrt{r/8n}$

$$\mathcal{P}_T[Y] \approx UV^\top$$

iii. $\|\mathcal{P}_{T^\perp}[Y]\| \leq 1/2$

$\mathcal{P}_{T^\perp}[Y]$ is small

$$\begin{aligned} \langle A, B \rangle &= \text{tr}[A^\top B] \\ &= \text{tr}[A^\top (P + P^\perp) B] \\ &= \text{tr}[A^\top P P B] + \text{tr}[A^\top P^\perp P^\perp B] \\ &= \langle PA, PB \rangle + \langle P^\perp A, P^\perp B \rangle \end{aligned}$$

$$\begin{aligned} \langle Z, UV^\top + U_\perp V_\perp^\top \rangle &= \langle Z, UV^\top + U_\perp V_\perp^\top - Y \rangle \\ &= \langle \mathcal{P}_T[Z], \mathcal{P}_T[UV^\top + U_\perp V_\perp^\top - Y] \rangle + \langle \mathcal{P}_{T^\perp}[Z], \mathcal{P}_{T^\perp}[UV^\top + U_\perp V_\perp^\top - Y] \rangle \\ &= \langle \mathcal{P}_T[Z], UV^\top - \mathcal{P}_T[Y] \rangle + \langle \mathcal{P}_{T^\perp}[Z], U_\perp V_\perp^\top - \mathcal{P}_{T^\perp}[Y] \rangle \\ &= \langle \mathcal{P}_T[Z], UV^\top - \mathcal{P}_T[Y] \rangle + \|\mathcal{P}_{T^\perp}[Z]\|_* - \langle \mathcal{P}_{T^\perp}[Z], \mathcal{P}_{T^\perp}[Y] \rangle \end{aligned}$$

duality

Matrix Completion

Suppose there is a **helper matrix** Y that satisfies the following conditions:

- i. Y is supported on Ω $\langle Y, Z \rangle = 0$
- ii. $\|\mathcal{P}_T[Y] - UV^\top\|_F \leq \sqrt{r/8n}$ $\mathcal{P}_T[Y] \approx UV^\top$
- iii. $\|\mathcal{P}_{T^\perp}[Y]\| \leq 1/2$ $\mathcal{P}_{T^\perp}[Y]$ is small

$$\langle Z, UV^\top + U_\perp V_\perp^\top \rangle = \langle \mathcal{P}_T[Z], UV^\top - \mathcal{P}_T[Y] \rangle + \|\mathcal{P}_{T^\perp}[Z]\|_* - \langle \mathcal{P}_{T^\perp}[Z], \mathcal{P}_{T^\perp}[Y] \rangle$$

$$\text{(by ii.)} \quad \geq -\sqrt{\frac{r}{8n}} \|\mathcal{P}_T[Z]\|_F + \|\mathcal{P}_{T^\perp}[Z]\|_* - \langle \mathcal{P}_{T^\perp}[Z], \mathcal{P}_{T^\perp}[Y] \rangle$$

$$\text{(by iii. and duality)} \quad \geq -\sqrt{\frac{r}{8n}} \|\mathcal{P}_T[Z]\|_F + \|\mathcal{P}_{T^\perp}[Z]\|_* - \frac{1}{2} \|\mathcal{P}_{T^\perp}[Z]\|_* = -\sqrt{\frac{r}{8n}} \|\mathcal{P}_T[Z]\|_F + \frac{1}{2} \|\mathcal{P}_{T^\perp}[Z]\|_*$$

- Thus, we need to show that $-\sqrt{r/8n} \|\mathcal{P}_T[Z]\|_F + (1/2) \|\mathcal{P}_{T^\perp}[Z]\|_* > 0$

Matrix Completion

Let R_Ω be a linear operator that zeros out all the entries of a matrix except those in Ω

$$\mathcal{R}_\Omega[Z] = \sum_{(i,j) \in \Omega} \langle \mathbf{e}_i \mathbf{e}_j^\top, Z \rangle \mathbf{e}_i \mathbf{e}_j^\top$$

Lemma. If Ω is chosen uniformly at random and $m \geq \Omega(nr \log n)$, then with high probability

$$\frac{n^2}{m} \left\| \mathcal{P}_T \mathcal{R}_\Omega \mathcal{P}_T - \frac{m}{n^2} \mathcal{P}_T \right\| < \frac{1}{2}$$

Matrix Completion

Lemma. If Ω is chosen uniformly at random and $m \geq rn \log n$, then with high probability for any Z supported in Ω^c we have

$$\|\mathcal{P}_{T^\perp}[Z]\|_* > \sqrt{\frac{r}{2n}} \|\mathcal{P}_T[Z]\|_F$$

Proof.

- The previous lemma gives that with high probability,

$$\left\| \mathcal{P}_T \mathcal{R}_\Omega \mathcal{P}_T - \frac{m}{n^2} \mathcal{P}_T \right\| < \frac{m}{2n^2}$$

- By Cauchy-Schwarz,

$$\left| \left\langle Z, \mathcal{P}_T \mathcal{R}_\Omega \mathcal{P}_T[Z] - \frac{m}{n^2} \mathcal{P}_T[Z] \right\rangle \right| \leq \left\| \mathcal{P}_T \mathcal{R}_\Omega \mathcal{P}_T - \frac{m}{n^2} \mathcal{P}_T \right\| \cdot \|Z\|_F^2$$

$$\left\langle Z, \mathcal{P}_T \mathcal{R}_\Omega \mathcal{P}_T[Z] - \frac{m}{n^2} \mathcal{P}_T[Z] \right\rangle \geq -\frac{m}{2n^2} \|Z\|_F^2$$

Matrix Completion

$$\left\langle Z, \mathcal{P}_T \mathcal{R}_\Omega \mathcal{P}_T[Z] - \frac{m}{n^2} \mathcal{P}_T[Z] \right\rangle = \langle Z, \mathcal{P}_T \mathcal{R}_\Omega \mathcal{P}_T[Z] \rangle - \frac{m}{n^2} \langle Z, \mathcal{P}_T[Z] \rangle = \langle Z, \mathcal{P}_T \mathcal{R}_\Omega \mathcal{P}_T[Z] \rangle - \frac{m}{n^2} \|\mathcal{P}_T[Z]\|_F^2$$

- Thus, we have

$$\begin{aligned} \langle Z, \mathcal{P}_T \mathcal{R}_\Omega \mathcal{P}_T[Z] \rangle &\geq \frac{m}{n^2} \|\mathcal{P}_T[Z]\|_F^2 - \frac{m}{2n^2} \|Z\|_F^2 \\ &= \frac{m}{n^2} \|\mathcal{P}_T[Z]\|_F^2 - \frac{m}{2n^2} \|\mathcal{P}_T[Z] + \mathcal{P}_{T^\perp}[Z]\|_F^2 \\ &= \frac{m}{2n^2} \|\mathcal{P}_T[Z]\|_F^2 - \frac{m}{2n^2} \|\mathcal{P}_{T^\perp}[Z]\|_F^2 \end{aligned}$$

- For LHS,

$$\begin{aligned} \langle Z, \mathcal{P}_T \mathcal{R}_\Omega \mathcal{P}_T[Z] \rangle &= \langle \mathcal{P}_T[Z], \mathcal{R}_\Omega \mathcal{P}_T[Z] \rangle \\ &= \langle \mathcal{R}_\Omega \mathcal{P}_T[Z], \mathcal{R}_\Omega \mathcal{P}_T[Z] \rangle = \|\mathcal{R}_\Omega \mathcal{P}_T[Z]\|_F^2 \\ &= \|\mathcal{R}_\Omega(Z - \mathcal{P}_{T^\perp}[Z])\|_F^2 = \|\mathcal{R}_\Omega \mathcal{P}_{T^\perp}[Z]\|_F^2 \\ &\leq \|\mathcal{P}_{T^\perp}[Z]\|_F^2 \end{aligned}$$

Matrix Completion

$$\|\mathcal{P}_{T^\perp}[Z]\|_F^2 \geq \frac{m}{2n^2} \|\mathcal{P}_T[Z]\|_F^2 - \frac{m}{2n^2} \|\mathcal{P}_{T^\perp}[Z]\|_F^2$$

$$\left(1 + \frac{m}{2n^2}\right) \|\mathcal{P}_{T^\perp}[Z]\|_F^2 \geq \frac{m}{2n^2} \|\mathcal{P}_T[Z]\|_F^2$$

$$\|\mathcal{P}_{T^\perp}[Z]\|_F^2 \geq \frac{m}{m + 2n^2} \|\mathcal{P}_T[Z]\|_F^2 \geq \frac{m}{4n^2} \|\mathcal{P}_T[Z]\|_F^2$$

- Note that $\|\mathcal{P}_{T^\perp}[Z]\|_*^2 \geq \|\mathcal{P}_{T^\perp}[Z]\|_F^2$
- Thus,

$$\|\mathcal{P}_{T^\perp}[Z]\|_* \geq \sqrt{\frac{m}{2n}} \|\mathcal{P}_T[Z]\|_F$$



Matrix Completion

Let $\mathcal{R}_\Omega$ be a linear operator that zeros out all the entries of a matrix except those in Ω

$$\mathcal{R}_\Omega[Z] = \sum_{(i,j) \in \Omega} \langle \mathbf{e}_i \mathbf{e}_j^\top, Z \rangle \mathbf{e}_i \mathbf{e}_j^\top$$

Lemma. If Ω is chosen uniformly at random and $m \geq \Omega(nr \log n)$, then with high probability

$$\frac{n^2}{m} \left\| \mathcal{P}_T \mathcal{R}_\Omega \mathcal{P}_T - \frac{m}{n^2} \mathcal{P}_T \right\| < \frac{1}{2}$$

- The operator norm of a linear operator is defined as $\|\mathcal{T}\| = \max_{\|Z\|_F \leq 1} \|\mathcal{T}[Z]\|_F$
- In expectation, $\mathbb{E}[\mathcal{R}_\Omega(Z)] = \frac{m}{n^2} Z$, that is, $\mathbb{E}[\mathcal{R}_\Omega] = \frac{m}{n^2} \text{Id}$

$$\mathbb{E}[\mathcal{P}_T \mathcal{R}_\Omega \mathcal{P}_T] = \mathcal{P}_T \mathbb{E}[\mathcal{R}_\Omega] \mathcal{P}_T = \frac{m}{n^2} \mathcal{P}_T$$

$$\mathcal{R}_\Omega \mathcal{P}_T[Z] = \sum_{(i,j) \in \Omega} \langle \mathbf{e}_i \mathbf{e}_j^\top, \mathcal{P}_T[Z] \rangle \mathbf{e}_i \mathbf{e}_j^\top = \sum_{(i,j) \in \Omega} \langle \mathcal{P}_T[\mathbf{e}_i \mathbf{e}_j^\top], Z \rangle \mathbf{e}_i \mathbf{e}_j^\top$$

Matrix Completion

Let R_Ω be a linear operator that zeros out all the entries of a matrix except those in Ω

$$\mathcal{R}_\Omega[Z] = \sum_{(i,j) \in \Omega} \langle \mathbf{e}_i \mathbf{e}_j^\top, Z \rangle \mathbf{e}_i \mathbf{e}_j^\top$$

Lemma. If Ω is chosen uniformly at random and $m \geq \Omega(nr \log n)$, then with high probability

$$\frac{n^2}{m} \left\| \mathcal{P}_T \mathcal{R}_\Omega \mathcal{P}_T - \frac{m}{n^2} \mathcal{P}_T \right\| < \frac{1}{2}$$

- The operator norm of a linear operator is defined as $\|\mathcal{T}\| = \max_{\|Z\|_F \leq 1} \|\mathcal{T}[Z]\|_F$
- $\mathbb{E}[\mathcal{P}_T \mathcal{R}_\Omega \mathcal{P}_T] = \frac{m}{n^2} \mathcal{P}_T$

$$\mathcal{P}_T \mathcal{R}_\Omega \mathcal{P}_T[Z] = \sum_{(i,j) \in \Omega} \underbrace{\langle \mathcal{P}_T[\mathbf{e}_i \mathbf{e}_j^\top], Z \rangle \mathcal{P}_T[\mathbf{e}_i \mathbf{e}_j^\top]}_{\mathcal{T}_{ij}[Z]}$$

Matrix Completion

Lemma. If Ω is chosen uniformly at random and $m \geq \Omega(nr \log n)$, then with high probability

$$\frac{n^2}{m} \left\| \mathcal{P}_T \mathcal{R}_\Omega \mathcal{P}_T - \frac{m}{n^2} \mathcal{P}_T \right\| < \frac{1}{2}$$

- Define linear operators $\mathcal{T}_{ij}$ for every $i, j \in [n]$:

$$\mathcal{T}_{ij}[Z] = \langle \mathcal{P}_T[\mathbf{e}_i \mathbf{e}_j^\top], Z \rangle \mathcal{P}_T[\mathbf{e}_i \mathbf{e}_j^\top]$$

- $\mathcal{P}_T \mathcal{R}_\Omega \mathcal{P}_T = \sum_{k=1}^m \mathcal{T}_{i_k j_k}$

- $\mathcal{T}_{ij}$ can be viewed as an $n^2 \times n^2$ matrix of rank-1: $\text{vec}(\mathcal{P}_T[\mathbf{e}_i \mathbf{e}_j^\top]) \text{vec}(\mathcal{P}_T[\mathbf{e}_i \mathbf{e}_j^\top])^\top$

- $\|\mathcal{T}_{ij}\| = \|\mathcal{P}_T[\mathbf{e}_i \mathbf{e}_j^\top]\|_F^2$

- $\mathbb{E}[\mathcal{T}_{i_k j_k}] = \frac{1}{n^2} \mathcal{P}_T$

- $\|\mathcal{T}_{i_k j_k} - \mathbb{E}[\mathcal{T}_{i_k j_k}]\| \leq \max\{\|\mathcal{P}_T[\mathbf{e}_{i_k} \mathbf{e}_{j_k}^\top]\|_F^2, 1/n^2\}$

For PSD matrices A, B ,
 $\|A - B\| \leq \max\{\|A\|, \|B\|\}$

Matrix Completion

- $\|\mathcal{J}_{i_k j_k} - \mathbb{E}[\mathcal{J}_{i_k j_k}]\| \leq \max\left\{\|\mathcal{P}_T[\mathbf{e}_{i_k} \mathbf{e}_{j_k}^\top]\|_F^2, 1/n^2\right\}$
- Recall that $\mathcal{P}_T[Z] = P_U Z + Z P_V - P_U Z P_V$

$$\|\mathcal{P}_T[\mathbf{e}_i \mathbf{e}_j^\top]\|_F^2 = \langle \mathcal{P}_T[\mathbf{e}_i \mathbf{e}_j^\top], \mathbf{e}_i \mathbf{e}_j^\top \rangle = \|P_U \mathbf{e}_i\|^2 + \|P_V \mathbf{e}_j\|^2 - \|P_U \mathbf{e}_i\|^2 \|P_V \mathbf{e}_j\|^2$$

- By the coherence assumption, $\|P_U \mathbf{e}_i\|^2, \|P_V \mathbf{e}_j\|^2 \leq \mu_0 r/n$

$$\|\mathcal{P}_T[\mathbf{e}_i \mathbf{e}_j^\top]\|_F^2 \leq \frac{2\mu_0 r}{n}$$

- Thus,

$$\|\mathcal{J}_{i_k j_k} - \mathbb{E}[\mathcal{J}_{i_k j_k}]\| \leq \max\left\{\frac{2\mu_0 r}{n}, \frac{1}{n^2}\right\} = \frac{2\mu_0 r}{n}$$

Matrix Completion

- $\|\mathcal{T}_{i_k j_k} - \mathbb{E}[\mathcal{T}_{i_k j_k}]\|$

$$\begin{aligned} \left\| \mathbb{E} \left[(\mathcal{T}_{i_k j_k} - \mathbb{E}[\mathcal{T}_{i_k j_k}])^2 \right] \right\| &= \left\| \mathbb{E} \left[\left(\mathcal{T}_{i_k j_k} - \frac{1}{n^2} \mathcal{P}_T \right)^2 \right] \right\| = \left\| \mathbb{E}[\mathcal{T}_{i_k j_k}^2] - \frac{1}{n^4} \mathcal{P}_T \right\| \\ &\leq \max \left\{ \left\| \mathbb{E}[\mathcal{T}_{i_k j_k}^2] \right\|, \frac{1}{n^4} \right\} \end{aligned}$$

- $\mathcal{T}_{ij} = \text{vec}(\mathcal{P}_T[\mathbf{e}_i \mathbf{e}_j^\top]) \text{vec}(\mathcal{P}_T[\mathbf{e}_i \mathbf{e}_j^\top])^\top$, so $\mathcal{T}_{ij}^2 = \|\mathcal{P}_T[\mathbf{e}_i \mathbf{e}_j^\top]\|_F^2 \cdot \mathcal{T}_{ij} \leq \frac{2\mu_0 r}{n} \mathcal{T}_{ij}$

$$\left\| \mathbb{E}[\mathcal{T}_{i_k j_k}^2] \right\| \leq \frac{2\mu_0 r}{n} \left\| \mathbb{E}[\mathcal{T}_{i_k j_k}] \right\| = \frac{2\mu_0 r}{n^3}$$

- Thus,

$$\left\| \mathbb{E} \left[(\mathcal{T}_{i_k j_k} - \mathbb{E}[\mathcal{T}_{i_k j_k}])^2 \right] \right\| \leq \max \left\{ \frac{2\mu_0 r}{n^3}, \frac{1}{n^4} \right\} = \frac{2\mu_0 r}{n^3}$$

Matrix Completion

$$\Pr \left[\left\| \sum_k Z_k \right\| \geq t \right] \leq (n_1 + n_2) \exp \left(-\frac{t^2}{2(v + Lt/3)} \right)$$

- $Z_k = \mathcal{J}_{i_k j_k} - \mathbb{E}[\mathcal{J}_{i_k j_k}]$
- $L = 2\mu_0 r/n, v = 2\mu_0 r/n^3$
- $t = m/(2n^2)$

$$\begin{aligned} \Pr \left[\left\| \sum_k (\mathcal{J}_{i_k j_k} - \mathbb{E}[\mathcal{J}_{i_k j_k}]) \right\| \geq \frac{m}{2n^2} \right] &\leq 2n^2 \exp \left(-\frac{m^2/n^4}{\mu_0 r \left(\frac{1}{n^3} + m/n^3 \right)} \right) \\ &= 2n^2 \exp \left(-\frac{m}{\mu_0 r n} \right) \leq \frac{1}{\text{poly}(n)} \end{aligned}$$

if $m \geq \Omega(\mu_0 r n \log n)$



Matrix Completion

It remains to show the existence of the **helper matrix** Y such that:

- i. Y is supported on Ω
- ii. $\|\mathcal{P}_T[Y] - UV^\top\|_F \leq \sqrt{r/8n}$
- iii. $\|\mathcal{P}_{T^\perp}[Y]\| \leq 1/2$

High-level strategy:

- Break up the observations Ω into disjoint subsets $\Omega_1, \dots, \Omega_p$ for $p = \log n$
- Use each subset of observations to construct Y iteratively:

$$\begin{aligned} Y_0 &= 0, & W_0 &= UV^\top \\ Y_{i+1} &= Y_i + \frac{n^2}{m'} \mathcal{R}_{\Omega_{i+1}}[W_i], & W_{i+1} &= UV^\top - \mathcal{P}_T[Y_{i+1}] \end{aligned}$$

- $\frac{n^2}{m'} \mathbb{E}[\mathcal{R}_{\Omega_{i+1}}] = \text{Id} \implies Y_{i+1} - Y_i$ is an unbiased estimator for W_i

Quantum
golfing



Matrix Completion

Quantum golfing

Quantum State Tomography via Compressed Sensing

[David Gross](#)¹, [Yi-Kai Liu](#)², [Steven T. Flammia](#)³, [Stephen Becker](#)⁴, and [Jens Eisert](#)⁵

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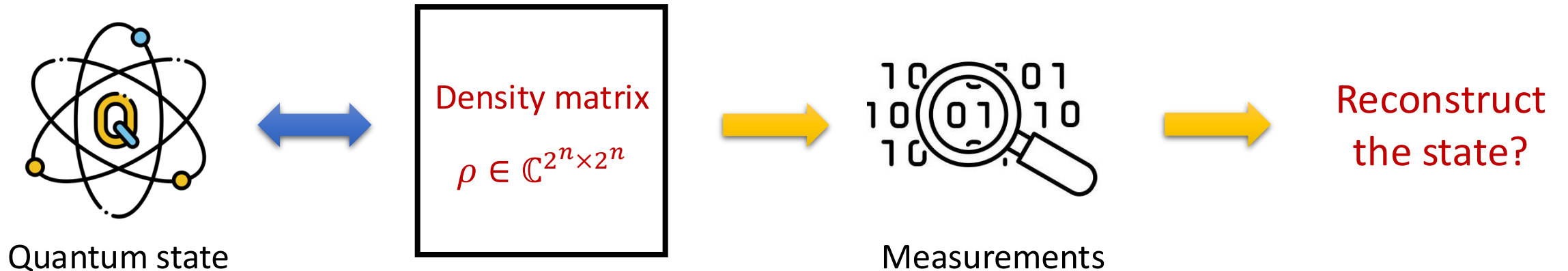
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Matrix Completion

High-level strategy:

- Break up the observations Ω into disjoint subsets $\Omega_1, \dots, \Omega_p$ for $p = \log n$
- Use each subset of observations to construct Y iteratively:

$$\begin{aligned} Y_0 &= 0, & W_0 &= UV^\top \\ Y_{i+1} &= Y_i + \frac{n^2}{m'} \mathcal{R}_{\Omega_{i+1}}[W_i], & W_{i+1} &= UV^\top - \mathcal{P}_T[Y_{i+1}] \end{aligned}$$

- $Y = Y_p$ is supported on Ω
- $\mathcal{P}_T[W_i] = W_i$ for all $i \in \{0, 1, \dots, p\}$

Matrix Completion

$$\begin{aligned}
 \|W_i\|_F &= \|\mathcal{P}_T[Y_i] - UV^\top\|_F = \left\| \mathcal{P}_T[Y_{i-1}] + \frac{n^2}{m'} \mathcal{P}_T \mathcal{R}_{\Omega_i}[W_{i-1}] - UV^\top \right\|_F \\
 &= \left\| \frac{n^2}{m'} \mathcal{P}_T \mathcal{R}_{\Omega_i}[W_{i-1}] - W_{i-1} \right\|_F \\
 &= \left\| \frac{n^2}{m'} \mathcal{P}_T \mathcal{R}_{\Omega_i} \mathcal{P}_T[W_{i-1}] - \mathcal{P}_T[W_{i-1}] \right\|_F \\
 &\leq \frac{n^2}{m'} \left\| \mathcal{P}_T \mathcal{R}_{\Omega_i} \mathcal{P}_T - \frac{m'}{n^2} \mathcal{P}_T \right\| \|W_{i-1}\|_F \\
 &\leq \frac{1}{2} \|W_{i-1}\|_F
 \end{aligned}$$

- The Frobenius norm of the remainder decreases geometrically
- Thus, Y satisfies $\|\mathcal{P}_T[Y] - UV^\top\|_F \leq \sqrt{r/8n}$ for $p = \log n$

Matrix Completion

$$Y_0 = 0, \quad W_0 = UV^\top$$

$$Y_{i+1} = Y_i + \frac{n^2}{m'} \mathcal{R}_{\Omega_{i+1}}[W_i], \quad W_{i+1} = UV^\top - \mathcal{P}_T[Y_{i+1}]$$

- $\|W_{i+1}\|_F \leq \frac{1}{2} \|W_i\|_F$
- It remains to show that $\|\mathcal{P}_{T^\perp}[Y]\| \leq 1/2$
- Let $\Delta_i Y := Y_i - Y_{i-1} = \frac{n^2}{m'} \mathcal{R}_{\Omega_i}[W_{i-1}]$
- $W_i \in T$, but $\mathcal{R}_{\Omega_i}[W_{i-1}]$ may not. We have $\Delta_i Y = \mathcal{P}_T[\Delta_i Y] + \mathcal{P}_{T^\perp}[\Delta_i Y]$ and

$$\mathcal{P}_{T^\perp}[\Delta_i Y] = \mathcal{P}_{T^\perp} \left[\frac{n^2}{m'} \mathcal{R}_{\Omega_i}[W_{i-1}] \right] = \mathcal{P}_{T^\perp} \left[\underbrace{\left(\frac{n^2}{m'} \mathcal{R}_{\Omega_i} - \text{Id} \right)}_{\text{sampling error}} [W_{i-1}] \right]$$

- Recall that $\mathbb{E} \left[\frac{n^2}{m'} \mathcal{R}_{\Omega_i} \right] = \text{Id}$

$$\mathcal{P}_{T^\perp}[W_{i-1}] = 0$$

Matrix Completion

$$\begin{aligned}\|\mathcal{P}_{T^\perp}[Y]\| &= \left\| \mathcal{P}_{T^\perp} \left[\sum_{i=1}^p \Delta_i Y \right] \right\| \leq \sum_{i=1}^p \left\| \mathcal{P}_{T^\perp} \left[\left(\frac{n^2}{m'} \mathcal{R}_{\Omega_i} - \text{Id} \right) [W_{i-1}] \right] \right\| \\ &\leq \sum_{i=1}^p \left\| \left(\frac{n^2}{m'} \mathcal{R}_{\Omega_i} - \text{Id} \right) [W_{i-1}] \right\|\end{aligned}$$

The second step follows because

$$\begin{aligned}\mathcal{P}_{T^\perp}[Z] &= (I - UU^\top)Z(I - VV^\top) \\ \|(I - UU^\top)Z(I - VV^\top)\| &\leq \|I - UU^\top\| \cdot \|Z\| \cdot \|I - VV^\top\| = \|Z\|\end{aligned}$$

- This is not true for a general projector on a linear subspace of **matrices**
 - **Counterexample:**

$$\begin{array}{llll} M = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} & \mathcal{P}_T \left[\begin{pmatrix} a & b \\ c & d \end{pmatrix} \right] = \begin{pmatrix} a & b \\ c & 0 \end{pmatrix} & A = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} & \mathcal{P}_T[A] = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix} \\ U, V = \mathbf{e}_1 & \mathcal{P}_{T^\perp} \left[\begin{pmatrix} a & b \\ c & d \end{pmatrix} \right] = \begin{pmatrix} 0 & 0 \\ 0 & d \end{pmatrix} & \|A\| = \sqrt{2} & \|\mathcal{P}_T[A]\| = \frac{1 + \sqrt{5}}{2} > \sqrt{2} \end{array}$$

Matrix Completion

$$\|\mathcal{P}_{T^\perp}[Y]\| \leq \sum_{i=1}^p \left\| \left(\frac{n^2}{m'} \mathcal{R}_{\Omega_i} - \text{Id} \right) [W_{i-1}] \right\|$$

Estimate 1:

- For a fixed matrix Z independent of Ω_i , by matrix Bernstein inequality, with probability $1 - \eta$

$$\left\| \left(\frac{n^2}{m'} \mathcal{R}_{\Omega_i} - \text{Id} \right) [Z] \right\| \leq \sqrt{\frac{n^3 \log(n/\eta)}{m'}} \|Z\|_{\max},$$

provided $m' \geq \Omega(rn \log(n/\eta))$

$$\|\mathcal{P}_{T^\perp}[Y]\| \leq \sum_{i=1}^p \left\| \left(\frac{n^2}{m'} \mathcal{R}_{\Omega_i} - \text{Id} \right) [W_{i-1}] \right\| \leq \sqrt{\frac{n^3 \log(n/\eta)}{m'}} \sum_{i=1}^p \|W_{i-1}\|_{\max}$$

Matrix Completion

Proof of Estimate 1:

$$\left(\frac{n^2}{m'} \mathcal{R}_{\Omega_i} - \text{Id}\right)[Z] = \frac{1}{m'} \sum_{k=1}^{m'} \underbrace{(n^2 Z_{i_k j_k} E_{i_k j_k} - Z)}_{X_k}$$

- $\mathbb{E}[X_k] = 0$ and $\|X_k\| \leq 2n^2 \|Z\|_{\max}$

$$\begin{aligned} \mathbb{E}[X_k X_k^\top] &= n^4 \mathbb{E}[Z_{i_k j_k}^2 E_{i_k i_k}] - n^2 \mathbb{E}[Z_{i_k j_k} E_{i_k j_k}] Z^\top - n^2 Z \mathbb{E}[Z_{i_k j_k} E_{j_k i_k}] + Z Z^\top \\ &= n^2 \sum_{ab} Z_{ab}^2 E_{ab} - Z Z^\top \preceq n^3 \|Z\|_{\max}^2 \cdot I \end{aligned}$$

- Matrix Bernstein inequality gives that

$$\Pr \left[\left\| \frac{1}{m'} \sum_k X_k \right\| \geq t \right] \leq n \exp \left(- \frac{t^2}{2(n^3 \|Z\|_{\max}^2 / m' + 2n^2 t / (3m'))} \right)$$

Matrix Completion

Proof of Estimate 1:

- Matrix Bernstein inequality gives that

$$\Pr \left[\left\| \frac{1}{m'} \sum_k X_k \right\| \geq t \right] \leq n \exp \left(- \frac{t^2}{2(n^3 \|Z\|_{\max}^2 / m' + 2n^2 t / (3m'))} \right)$$

if

$$t \geq \sqrt{\frac{n^3 \log(n/\eta)}{m'}} \|Z\|_{\max}$$



Matrix Completion

Estimate 2:

- For a fixed matrix $Z \in T$ independent of Ω_i , with probability $1 - \eta$,

$$\left\| Z - \frac{n^2}{m'} \mathcal{P}_T \mathcal{R}_{\Omega_i} [Z] \right\|_{\max} \leq \frac{1}{2} \|Z\|_{\max}$$

provided $m' \geq \Omega(\mu_0 r n \log(1/\eta))$

- Recall

$$Y_0 = 0,$$

$$W_0 = UV^\top$$

$$Y_{i+1} = Y_i + \frac{n^2}{m'} \mathcal{R}_{\Omega_{i+1}} [W_i], \quad W_{i+1} = UV^\top - \mathcal{P}_T [Y_{i+1}]$$

$$W_{i+1} - W_i = \mathcal{P}_T [Y_i - Y_{i+1}] = -\frac{n^2}{m'} \mathcal{P}_T \mathcal{R}_{\Omega_{i+1}} [W_i]$$

$$\|W_i\|_{\max} = \left\| W_{i-1} - \frac{n^2}{m'} \mathcal{P}_T \mathcal{R}_{\Omega_i} [W_{i-1}] \right\|_{\max} \leq \frac{1}{2} \|W_{i-1}\|_{\max} \leq \dots \leq 2^{-i} \|UV^\top\|_{\max}$$

Matrix Completion

$$\begin{aligned}
 \|\mathcal{P}_{T^\perp}[Y]\| &\leq \sum_{i=1}^p \left\| \left(\frac{n^2}{m'} \mathcal{R}_{\Omega_i} - \text{Id} \right) [W_{i-1}] \right\| \\
 &\leq \sqrt{\frac{n^3 \log(n/\eta)}{m'}} \sum_{i=1}^p \|W_{i-1}\|_{\max} && \text{(Estimate 1)} \\
 &\leq \sqrt{\frac{n^3 \log(n/\eta)}{m'}} \sum_{i=1}^p 2^{-(i-1)} \|UV^\top\|_{\max} && \text{(Estimate 2)} \\
 &\leq 2 \sqrt{\frac{n^3 \log(n/\eta)}{m'}} \frac{\mu_1 \sqrt{r}}{n} \leq \frac{1}{2} && \text{(Assumption)}
 \end{aligned}$$

- Provided $m' \geq \Omega(\mu_1^2 r n \log(n/\eta))$
- To make those estimates hold, we also need $m' \geq \Omega(\mu_0 r n \log(n/\eta))$

Matrix Completion

Let $n_1 = n_2 = n$ for simplicity and let $M = U\Sigma V^\top$ be the SVD with $U, V \in \mathbb{R}^{n \times r}, \Sigma \in \mathbb{R}^{r \times r}$

- We assume that $\mu(U), \mu(V) \leq \mu_0$
- We also assume that $\|UV^\top\|_{\max} \leq \mu_1 \sqrt{r}/n$

Theorem (Exact Recovery). Under these assumptions, with probability $1 - \delta$, the nuclear norm minimization

$$\hat{X} = \arg \min_X \|X\|_* \quad \text{s. t.} \quad X_{i_k j_k} = M_{i_k j_k} \quad \forall k \in [m]$$

will exactly recover M if

$$m \geq \Omega(\max\{\mu_1^2, \mu_0\} rn \log^2(n/\delta))$$

Matrix Completion

Proof of Estimate 2:

$$\begin{aligned} \left(\frac{n^2}{m'} \mathcal{P}_T \mathcal{R}_{\Omega_i} [Z] - Z \right)_{a,b} &= \frac{1}{m'} \sum_{k=1}^{m'} \left(n^2 Z_{i_k j_k} \langle \mathcal{P}_T [E_{i_k j_k}], E_{ab} \rangle - Z_{ab} \right) \\ &= \frac{1}{m'} \sum_{k=1}^{m'} \underbrace{\left(n^2 Z_{i_k j_k} \langle E_{i_k j_k}, \mathcal{P}_T [E_{ab}] \rangle - Z_{ab} \right)}_{\xi_k} \end{aligned}$$

- Recall that $\|\mathcal{P}_T [E_{ab}]\|_F^2 \leq \frac{2\mu_0 r}{n}$
- Thus, $|\xi_k| \leq n^2 \|Z\|_{\max} \|\mathcal{P}_T [E_{ab}]\|_F \|\mathcal{P}_T [E_{i_k j_k}]\|_F = 4\mu_0 r n \|Z\|_{\max}$

Matrix Completion

Proof of Estimate 2:

$$\left(\frac{n^2}{m'} \mathcal{P}_T \mathcal{R}_{\Omega_i} [Z] - Z \right)_{a,b} = \frac{1}{m'} \sum_{k=1}^{m'} \underbrace{(n^2 Z_{i_k j_k} \langle E_{i_k j_k}, \mathcal{P}_T [E_{ab}] \rangle - Z_{ab})}_{\xi_k}$$

- $\|\mathcal{P}_T [E_{ab}]\|_F^2 \leq \frac{2\mu_0 r}{n}$
- $|\xi_k| \leq 4\mu_0 r n \|Z\|_{\max}$
- Note that $\mathbb{E}[n^2 Z_{i_k j_k} \langle E_{i_k j_k}, \mathcal{P}_T [E_{ab}] \rangle] = Z_{ab}$
- Thus, $\mathbb{E}[\xi_k^2] \leq \mathbb{E}[(n^2 Z_{i_k j_k} \langle E_{i_k j_k}, \mathcal{P}_T [E_{ab}] \rangle)^2] \leq n^2 \|Z\|_{\max}^2 \|\mathcal{P}_T [E_{ab}]\|_F^2 \leq 2\mu_0 r n \|Z\|_{\max}^2$

Matrix Completion

Proof of Estimate 2:

$$\left(\frac{n^2}{m'} \mathcal{P}_T \mathcal{R}_{\Omega_i} [Z] - Z \right)_{a,b} = \frac{1}{m'} \sum_{k=1}^{m'} \xi_k$$

- $|\xi_k| \leq 4\mu_0 rn \|Z\|_{\max}$ and $\mathbb{E}[\xi_k^2] \leq 2\mu_0 rn \|Z\|_{\max}^2$
- By the scalar Bernstein inequality,

$$\Pr \left[\left| \sum_{k=1}^{m'} \frac{1}{m'} \xi_k \right| \geq t \right] \leq 2 \exp \left(- \frac{t^2/2}{2\mu_0 rn \|Z\|_{\max}^2 / m' + 4\mu_0 rn \|Z\|_{\max} t / (3m')} \right)$$

- If we take $t = \frac{1}{2} \|Z\|_{\max}$, then $\text{RHS} \leq \eta$ provided $m' \geq \Omega(\mu_0 rn \log(1/\eta))$

